Data Analytics Portfolio DS_02

Two end-to-end, client-ready analytics dashboards built with Python and Plotly Dash.

Projects

1. E-Commerce Sales Dashboard

Dataset: UCI E-Commerce Data — Kaggle

Analyses 541,909 transactions from a UK-based online retailer (2010-2011) to answer:

- Which products generate the most revenue?
- How do sales change over time?
- Which countries and categories are most profitable?
- Where should the business focus to grow faster?

Key Findings:

- Total revenue of GBP 9.75M across 2 years with 23.4% YoY growth
- November is consistently the peak month driven by holiday wholesale orders
- 82% of revenue comes from the UK -- international markets are largely untapped
- 34% of customers are "at risk" -- re-engagement campaign opportunity
- Saturday/Sunday orders near zero confirming a B2B wholesale customer base

---

2. Customer Churn & Retention Dashboard

Dataset: Telco Customer Churn — Kaggle

Analyses 7,043 telecom customers to answer:

- Why are customers leaving the platform?
- Which segments are most likely to churn?
- How long do customers typically stay active?
- What actions can improve retention?

Key Findings:

- Overall churn rate of 26.5% with USD 139K monthly revenue at risk
- Month-to-month contract customers churn at 42% vs 3% for two-year contracts
- Fibre optic customers churn at nearly 2x the rate of DSL customers
- Customers without OnlineSecurity or TechSupport churn significantly more
- The first 12 months are the highest-risk window for customer loss

---

Dashboards

Dashboard	File	Dataset
E-Commerce Sales	app.py	data.csv
Churn & Retention	churn_app.py	WA_Fn-UseC_-Telco-Customer-Churn.csv

---

Setup

1. Clone the repo

git clone https://github.com/YOUR_USERNAME/DS_02.git

cd DS_02

2. Install dependencies

pip install -r requirements.txt

3. Download datasets

- **E-Commerce:** https://www.kaggle.com/datasets/carrie1/ecommerce-data
  - Download data.csv and place it in the project folder
- **Churn:** https://www.kaggle.com/datasets/blastchar/telco-customer-churn
  - Download WA_Fn-UseC_-Telco-Customer-Churn.csv and place it in the project folder

4. Update file paths (if needed)

In app.py (bottom of file):

CSV = r"C:\path\to\your\data.csv"

In churn_app.py (bottom of file):

CSV = r"C:\path\to\your\WA_Fn-UseC_-Telco-Customer-Churn.csv"

5. Run the dashboards

E-Commerce dashboard

python app.py

Open http://127.0.0.1:8050

Churn dashboard (in a new terminal)

python churn_app.py

Open http://127.0.0.1:8050

Features

E-Commerce Dashboard (app.py)

- KPI cards: Revenue, Orders, Customers, AOV, Countries
- Monthly revenue trend (2010 vs 2011) with year filter
- Top 15 products by revenue
- Revenue by country (UK + international breakdown)
- Orders by day of week (reveals B2B weekday pattern)
- Revenue by product category
- RFM customer segmentation (Champions / Loyal / At Risk / Lost)
- International markets chart (ex-UK)
- Key insights + actionable recommendations panel

Churn & Retention Dashboard (churn_app.py)

- KPI cards: Customers, Churn rate, Retention, Avg tenure, Revenue at risk
- Overall churn donut chart
- Customer retention survival curve
- Churn by contract type (interactive filter)
- Churn by tenure band (interactive filter)
- Churn by payment method
- Churn by internet service type
- Senior vs non-senior segmentation
- Monthly charges distribution: retained vs churned
- Churn rate by charge band
- Random Forest feature importance (top churn predictors)
- Key insights + actionable recommendations panel

Tech Stack

Tool	Purpose
Python 3.8+	Core language
pandas	Data cleaning and aggregation

Tool	Purpose
numpy	Numerical operations
plotly	Interactive charts
Dash	Web dashboard framework
scikit-learn	Random Forest churn prediction model

Project Structure

```
DS_02/
|-- app.py                    # E-Commerce dashboard (all-in-one)
|-- churn_app.py              # Churn & Retention dashboard (all-in-one)
|-- requirements.txt          # Python dependencies
|-- README.md                 # This file
|-- data.csv                  # E-Commerce dataset (download from Kaggle)
|-- WA_Fn-UseC_-Telco-Customer-Churn.csv  # Churn dataset (download from Kaggle)
```

Skills Demonstrated

- Data cleaning and preparation (missing values, type casting, feature engineering)
- Exploratory data analysis (EDA)
- Revenue trend analysis and forecasting context
- Customer segmentation (RFM analysis)
- Cohort and retention analysis
- Churn prediction modelling (Random Forest)
- Interactive dashboard development
- Business insight generation and storytelling
- Client-ready data visualisation

Author

Boitumelo Data Analyst | Python | SQL | Power BI | Dash

Data Sources

- Chen, D. et al. (2012). *Online Retail Dataset*. UCI Machine Learning Repository via Kaggle.
- IBM Watson Analytics (2018). *Telco Customer Churn*. Kaggle.